

UCS615: Image Processing

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Digital Image Fundamentals:

Image perception - light, luminance, brightness, and contrast; Examples of fields that use digital image processing; Digital Image Fundamentals: A simple image formation model, image sampling and quantization, basic relationships between pixels, Types of Images: Binary, Grayscale, color; Color representation: Color models; Pseudo-color and Full-color image processing.

Image Enhancement and Restoration in Spatial and Frequency (Transform) domain:

Introduction to spatial and frequency (transform) domain, DFT (Discrete Fourier Transform), Properties of 2-D DFT; Image enhancement: Efficient algorithms for point processing in real-time, low-power systems, Neighbourhood processing, Histogram processing; Image Smoothing and Sharpening: Lightweight smoothing/sharpening filters for resource-constrained devices; Image restoration: degradation model; inverse filtering.

Feature Extraction:

Introduction to type of features; Boundary Feature Descriptors; Region Feature Descriptors; Principal Components as Feature Descriptors; Whole Edges: Canny, LOG, DOG, Hough Transform, Corners: Harris-Stephens corner detector; Whole Image Features: Optimizing SIFT for resource-constrained environments, LBP (Local Binary Pattern) and its variants.

Image Segmentation:

Morphological operation, Point-Line-Edge Detection; Thresholding; Segmentation by Region Growing; Segmentation by Region Splitting and Merging; Region Segmentation by Clustering; Region Segmentation by Graph-Cut; Texture Segmentation; Introduction to use of Motion in Segmentation.

Energy-efficient algorithms for application areas of Image Processing:

Image compression: JPEG compression; Huffman coding.

Image security: Watermarking, Steganography, Visual Cryptography.

Object Detection and Classification: Introduction to Neural Network, Convolutional Neural Networks (CNNs), Overview on use cases of CNNs in Image Processing, Study of prominent CNN architectures: AlexNet, ResNet, EfficientNet, etc.

Laboratory work: Demonstrate the use of Image Processing Toolbox on MATLAB/PYTHON to create interactive image processing applications like image enhancement, image compression, image segmentation, feature extraction etc.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Comprehend the need and usage of concepts of image processing.
2. Enhance the visual quality of given grey/color image using well known transformations and filters.
3. Apply and comprehend the role of feature extraction in Image Processing.
4. Demonstrate the use of image processing techniques to ideate innovative solutions to real-world problems .

Text Books:

1. Gonzalez C. R., Woods E. R., *Digital Image Processing*, Pearson Education, 4th ed.
2. Anil K. Jain, *Fundamentals of Digital Image Processing*, Prentice Hall, 1989.

Reference Books:

1. McAndrew A., *Introduction to Digital Image Processing with Matlab*, Thomson Course Technology (2004)
2. Low A., *Introductory Computer Vision and Image Processing*, McGraw-Hill (1991), 1sted.